Report on Resource Adequacy Plan (Generation) for Tamil Nadu (2025-26 to 2035-36)

April 2026

Government of India
Ministry of Power
Central Electricity Authority

Contents

Disclaimer

This Resource Adequacy study for Tamil Nadu has been conducted based on the data and inputs provided by Tamil Nadu. The findings, analysis and conclusions presented in this report are contingent upon the accuracy, completeness and timeliness of the information furnished by Tamil Nadu. Any discrepancies or limitations in the data may affect the outcomes of the study accordingly.

In accordance with the Resource Adequacy Guidelines dated 28th June 2023, each Distribution Licensee is mandated to prepare a Resource Adequacy Plan (RAP) for a 10-year horizon, referred to as the Long-Term Distribution Licensee Resource Adequacy Plan (LT-DRAP), which shall be vetted and validated by the Central Electricity Authority (CEA). CEA has facilitated this study and prepared the report solely to assist Tamil Nadu in fulfilling this requirement.

It is expressly stated that the responsibility for the implementation of the study's recommendations, ensuring the adequacy of electricity resources, and undertaking any related actions including financial implications, if any, rests entirely with Tamil Nadu.

Executive Summary

The Ministry of Power had notified the Electricity (Amendment) Rules in December 2022. As per Rule 16 of the Electricity (Amendment) Rules, the Ministry of Power has notified the Resource Adequacy (RA) Guidelines. According to these Guidelines, the Central Electricity Authority (CEA) is entrusted with the responsibility of preparing the Long-Term National Resource Adequacy Plan (LT-NRAP). Further, each Distribution Utility is required to carry out a Long-Term Distribution Licensee Resource Adequacy Plan (LT-DRAP) to reliably meet its peak electricity demand and electrical energy requirements.

As per the Resource Adequacy Guidelines dated 28th June 2023, each Distribution Licensee shall undertake a Resource Adequacy Plan (RAP) with a 10-year planning horizon, LT-DRAP, to meet its own peak electricity demand and electrical energy requirements. This plan shall be vetted/validated by the CEA to leverage the benefits of national-level optimization for the Distribution Licensees. The LT-DRAP shall be prepared by the Distribution Licensees on an annual rolling basis, factoring in the already contracted capacity and optimizing the requirement for additional capacity.

The Government of India has gazette-notified the Renewable Consumption Obligation (RCO) trajectory up to 2029-30 vide gazette notification dated 27th September 2025, which mandates that a specified portion of electrical energy consumption must be met from renewable energy sources.

To support the utility in fulfilling this requirement, CEA, in collaboration with the utility, initially conducted a Resource Adequacy (RA) study with a planning horizon up to FY 2034-35, based on the data received from the utility and in compliance with the RCO trajectory specified vide gazette notification dated 20th October 2023.

The electrical energy requirement and peak electricity demand for Tamil Nadu are increasing with a CAGR of 5.8% and 5.6% respectively, from 2025-26 to 2035-36 as forecasted by Tamil Nadu. For satisfying resource adequacy, i.e., meeting the electricity demand reliably and at an affordable cost, the utility needs to methodically plan its capacity expansion either by investing in generation capacity or by procuring power. In view of the reduction in cost of solar, wind and technology options like Battery Energy Storage Systems (BESS), planning for a long-term optimal generation capacity mix gains tremendous importance so that the future generation capacity mix is cost-effective as well as environmentally friendly.

A study has been carried out for Tamil Nadu, considering the existing contracted capacity and planned capacity addition. It was found that Tamil Nadu's likely contracted capacity may not be sufficient to meet the projected electricity demand. Further, the total unserved electrical energy in the year 2035-36 is expected to be about 56,294 MU, which is about 21.4% of the projected annual electrical energy requirement during the year 2035-36.

To find out the least cost option for generation capacity expansion for the period 2025-26 to 2035-36, a long-term study has been carried out with the objective to minimize the total system cost of generation, including the cost of anticipated future investments, while fulfilling all the technical constraints associated with various power generation technologies. Additionally, a reliability study has been carried out to determine the probability of unmet demand and hours, by implementing the variation in demand, variation in RE and forced outage of thermal generators (Coal/ Lignite), etc.

Generation capacity expansion pathways have been considered for the long-term study based on the yearly capacity addition plans of the utility, along with RCO constraints. The Renewable capacities have been assessed in view of adherence to the RCO trajectory gazette notified by the Ministry of Power, considering the fungibility among different sources.

The Resource adequacy studies have projected a likely optimal capacity mix for future years till 2035-36, which is able to meet the projected electrical demand reliably at every instance. As per the Resource Adequacy studies, the likely total projected contracted capacity for the year 2035-36 is around 1,22,498 MW which consists of 23,504 MW from Coal; 408 MW from Gas; 2675 MW from Nuclear; 1900 MW from Hydro; 20,181 MW from Wind; 36,156 MW from Solar; 14,761 MW from Distributed Renewable Energy (DRE) sources; 966 MW from Biomass; 11,575 MW/45300 MWh of Storage; 6264 MW/39782 MWh of PSP and 4107 MW of MTOA/STOA. This capacity shall be able to meet the projected electricity demand with prescribed reliability criteria.

1.0 Introduction

Ministry of Power has notified the Electricity (Amendment) Rules, 2022, in December 2022. Rule 16 (I) of the said rules stipulates that "*A guideline for assessment of resource adequacy during the generation planning stage (one year or beyond) as well as during the operational planning stage (up to one year) shall be issued by the Central Government in consultation with the Authority*". Accordingly, the Resource Adequacy Guidelines were notified in June 2023 by the Ministry of Power in consultation with the Central Electricity Authority.

Resource Adequacy is generally defined as a mechanism to ensure that there is an adequate supply of generation resources to serve expected demand reliably at least cost. A key aspect of resource adequacy planning is to ensure that adequate generation capacities are available, round-the-clock, to reliably serve demand under various scenarios. This naturally translates into the need for ensuring adequate reserve margin, which could cater to varying levels of demand and supply conditions in the grid. In the wake of high RE generation, it is important to understand the demand-supply situation in the grid precisely due to high seasonality and intermittency in RE generation. Resource Adequacy exercise may also help in the assessment of capacity requirements to be tied up or contracted on a long-term, medium-term, and short-term basis.

Further, the Ministry of Power vide notification dated 27th September 2025, had notified the RCO trajectory for the states/Discoms. Based on the trajectory specified, hydro, wind and other (solar, biomass, etc.) RCO quantum in million units (MUs) has been calculated to find additional quantum of renewable capacity that the states/Discoms have to contract in addition to their existing/planned capacity to meet their RCO targets.

To support the utility in fulfilling the Resource Adequacy Guidelines and complying with the Renewable Consumption Obligation (RCO) notification, CEA has carried out the RA study for Tamil Nadu based on inputs furnished by the Utility. The study recommends an optimal resource mix up to FY 2035-36, taking into account technical and financial parameters associated with various capacities. It aims to optimise long-term power procurement while ensuring resource adequacy to meet demand on a 24x7 basis, considering variations in demand, RE generation, and forced outages of thermal capacities. The study assesses the Planning Reserve Margin (PRM) required by the utility to account for the aforementioned uncertainties, ensuring that demand can be reliably met throughout the year.

Prior to this, CEA had conducted the RA study for the Utility up to FY 2034-35, based on inputs from Tamil Nadu and in accordance with the RPO trajectory specified in the Ministry of Power's gazette notification dated 20th October 2023.

2.0 Highlights of the Previous RA Study (up to FY 2034-35)

1. In the earlier Resource Adequacy (RA) study, financial year 2023-24 had been considered as the base year, and the study covered the period from 2024-25 to 2034-35. The fuel-wise contracted capacity as on 31st March, 2024:

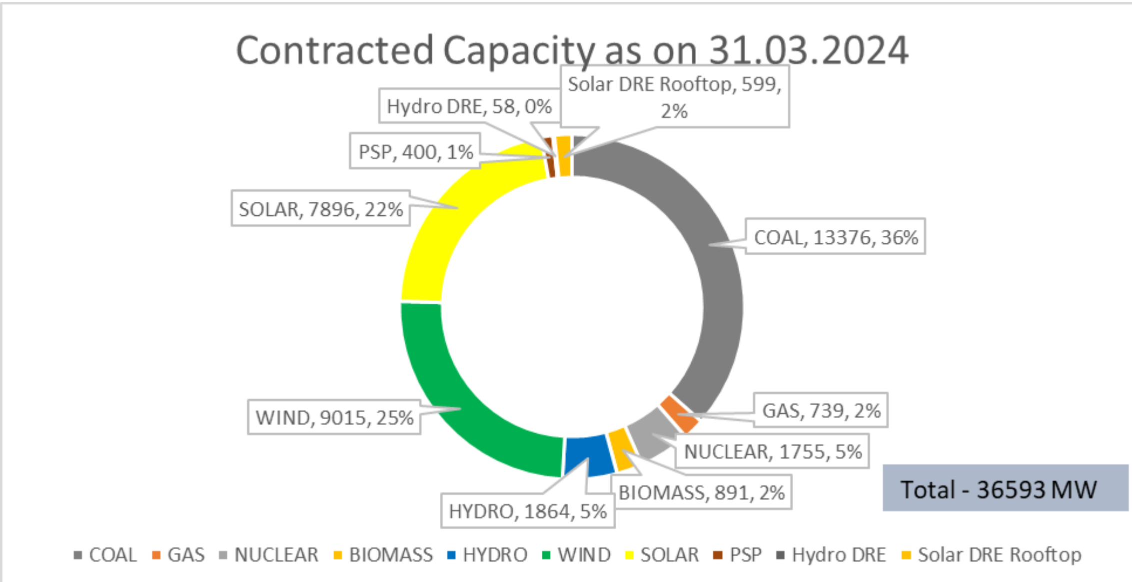


Figure 1: Fuel-wise Contract Capacity (in MW) as of March 2024

2. The hourly demand pattern for 2023-24 had been analysed and considered for the study.

3. The peak electricity demand and energy projections as per furnished by 20th EPS were lower than those projected in the Tamil Nadu. The studies have been carried out considering the demand projections as per Tamil Nadu, which are tabulated below:

Table 1: Electrical Energy requirement (MU) and Peak electricity demand (MW) projections

Year	**2023-24**	**2024-25**	**2025-26**	**2026-27**	**2027-28**	**2028-29**	**2029-30**	**2030-31**	**2031-32**	**2032-33**	**2033-34**	**2034-35**
Electrical Energy Requirement Projections (MU)	130223 (1,26,163*)	142441 (1,30,413*)	152678	162726	172679	182401	193232	203598	214582	225898	237559	249580
Year-on-Year Growth *(Electrical Energy)*		9.4%	7.2%	6.6%	6.1%	5.6%	5.9%	5.4%	5.4%	5.3%	5.2%	5.1%
Peak electricity demand Projections (MW)	19409 (19,045*)	20701 (20,784*)	21959	23314	24680	26046	27541	29027	30567	32159	33805	35507
Year-on-Year Growth (Peak electricity demand)		6.7%	6.1%	6.2%	5.9%	5.5%	5.7%	5.4%	5.3%	5.2%	5.1%	5.0%

*actual electrical energy requirement and peak electricity demand

4. The actual peak electricity demand and electrical energy requirement of the utility observed during 2024-25 is tabulated below:

Table 2: Actual Electrical Energy Requirement and Peak Electricity Demand in 2024-25

Year	Actual electrical energy requirement (in MU)	Actual peak electricity demand (in MW)
2024-25	1,30,413	20,784

5. To meet the above projected peak electricity demand and electrical energy requirement reliably, the source-wise projected capacity (in MW) as outlined in the previous report is tabulated below.

Table 3: Year-wise capacity projections (in MW)

	Coal	Gas	Nuclear	Biomass	Hydro	Wind	Solar	Solar DRE	Hydro DRE	Storage	STOA
2024/25	15291	739	1906	891	1864	10015	12496	1614	58	400	3274
2025/26	15141	739	1906	966	1884	11015	14796	2457	58	900	3996
2026/27	16461	739	3006	966	1884	12015	16796	3391	58	2441	2580
2027/28	16923	739	3006	966	1884	13015	18996	4419	58	4344	2674
2028/29	17588	739	1828	966	1884	14015	21496	5533	58	5581	3310
2029/30	18380	739	2828	966	1884	16515	24296	6778	58	7581	2408
2030/31	19380	739	2828	966	1884	18015	25796	7947	58	8581	2780
2031/32	20380	408	2828	966	1884	19515	27296	9224	58	9581	2855
2032/33	21380	408	2828	966	1884	21015	28796	10603	58	10581	2184
2033/34	22130	408	2828	966	1884	22515	30296	12089	58	12581	2084
2034/35	22497	408	2828	966	1884	24015	31796	13687	58	13581	2799

*****The STOA/MTOA value represents the peak requirement in MW, and it was recommended that this requirement may be met through power procurement from the market or through bilateral agreements.

6. The year-wise planned and additional capacity contract addition (in MW) for the above tabulated cumulative capacity is detailed below.

Table 4: Year wise capacity additions for Tamil Nadu (in MW)

FY	Coal		Solar		Wind		Yearly STOA
	Planned	Additional	Planned	Additional	Planned	Additional	Additional
2024/25	2120	0	4100	500	500	500	3274
2025/26	0	0	1800	500	500	500	3996
2026/27	1320	0	1500	500	500	500	2580
2027/28	462	0	1700	500	500	500	2674
2028/29	2082	1000	2000	500	500	500	3310
2029/30	500	1000	2300	500	2000	500	2408
2030/31	0	1000	1000	500	1000	500	2780
2031-32	0	1000	1000	500	1000	500	2855
2032/33	0	1000	1000	500	1000	500	2184
2033/34	0	1000	1000	500	1000	500	2084
2034/35	0	1000	1000	500	1000	500	2799
FY	Storage		DRE		Nuclear	Biomass	Hydro
	Planned	Additional	Planned	Additional	Planned	Planned	Planned
2024/25	0	0	300	715	152	0	0
2025/26	500	0	500	342	0	75	20
2026/27	0	1541	750	185	1100	0	0
2027/28	0	1903	750	277	0	0	0
2028/29	0	1237	750	364	0	0	0
2029/30	0	2000	1000	245	1000	0	0
2030/31	0	1000	1000	169	0	0	0
2031-32	0	1000	1000	277	0	0	0
2032/33	0	1000	1100	279	0	0	0
2033/34	1000	1000	1100	386	0	0	0
2034/35	0	1000	1100	498	0	0	0

3.0 RA Study for Tamil Nadu (from 2025-26 to 2035-36)

3.1 Present Power Scenario in Tamil Nadu

The power supply position for Tamil Nadu for 2021-22 to 2024-25 is given in the Table 5.

Table 5: Power Supply Position of Tamil Nadu

Power Supply Position						
Year	Energy required (MU)	Energy supplied (MU)	Gap (MU)	Peak Demand (MW)	Peak Met (MW)	Demand not met (MW)
2021-22	1,09,816	1,09,798	18	16,891	16,891	0
2022-23	1,14,798	1,14,722	77	17,729	17,729	0
2023-24	1,26,163	1,26,151	12	19,045	19,045	0
2024-25	1,30,413	1,30,408	5	20,784	20,784	0

As of March 2025, the total contracted capacity of Tamil Nadu is 37,220 MW, including 1003 MW DRE. Out of the total contracted capacity (CC), the share of non-fossil fuel-based CC is 64%. Fuel-wise contracted capacity as on 31st March, 2025, is given in Table 6 and Figure 2:

Table 6: Fuel-wise Contracted Capacity as on March 2025

Technology	**Contracted Capacity (MW)**	**%**
Coal	12808	35%
Gas	408	1%
Nuclear	1448	4%
Biomass	891	2%
Hydro	1866	5%
Wind	9331	25%
Solar	9006	24%
PSP	400	1%
Hydro DRE	58	0%
Solar DRE Rooftop	1003	3%
Total	**37220**	**100%**

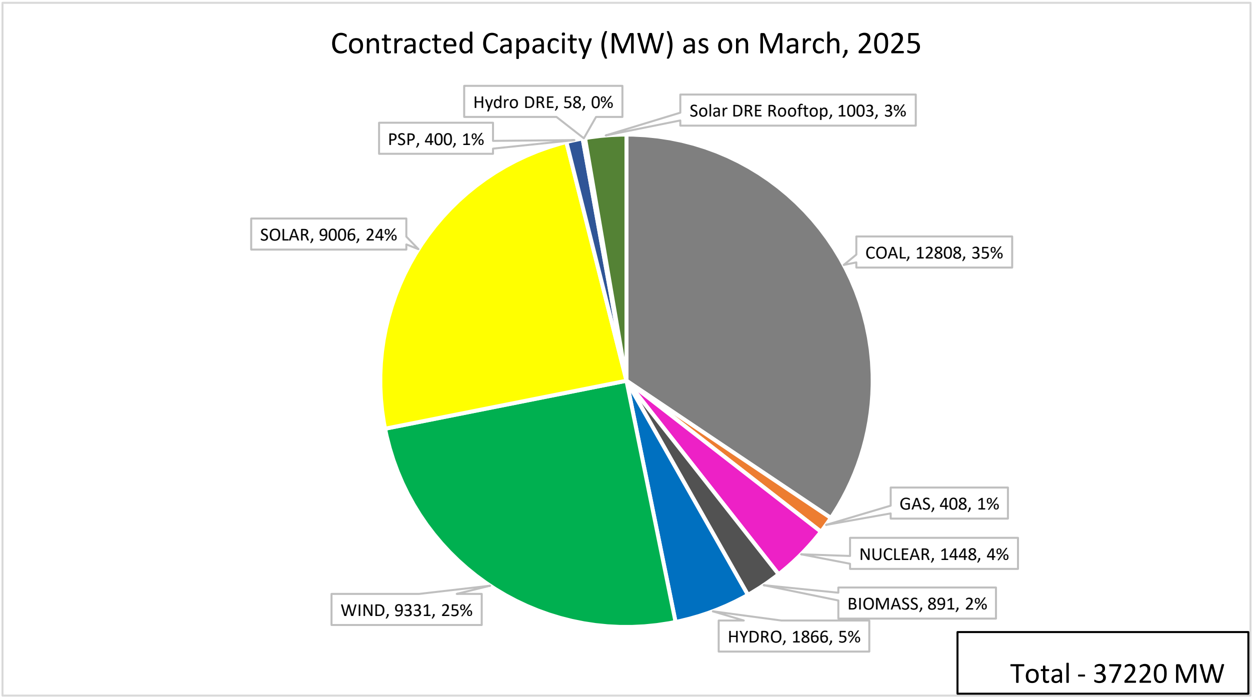


Figure 2: Fuel-wise Contracted Capacity (in MW) as on March, 2025

3.2 Demand Analysis of the FY 2024-25

The hourly demand pattern of 2024-25 was analyzed (as shown in Figure 3 and Figure 4), and it was observed that the peak demand for Tamil Nadu occurs in the months of March and April. The Demand Pattern of Tamil Nadu indicates that the state has a higher Day peak in comparison to Night Peak, with the daily peak mostly occurring during the 11:00 Hrs to 17:00 Hrs. Further, the demand pattern remains almost flat during the noon hours.

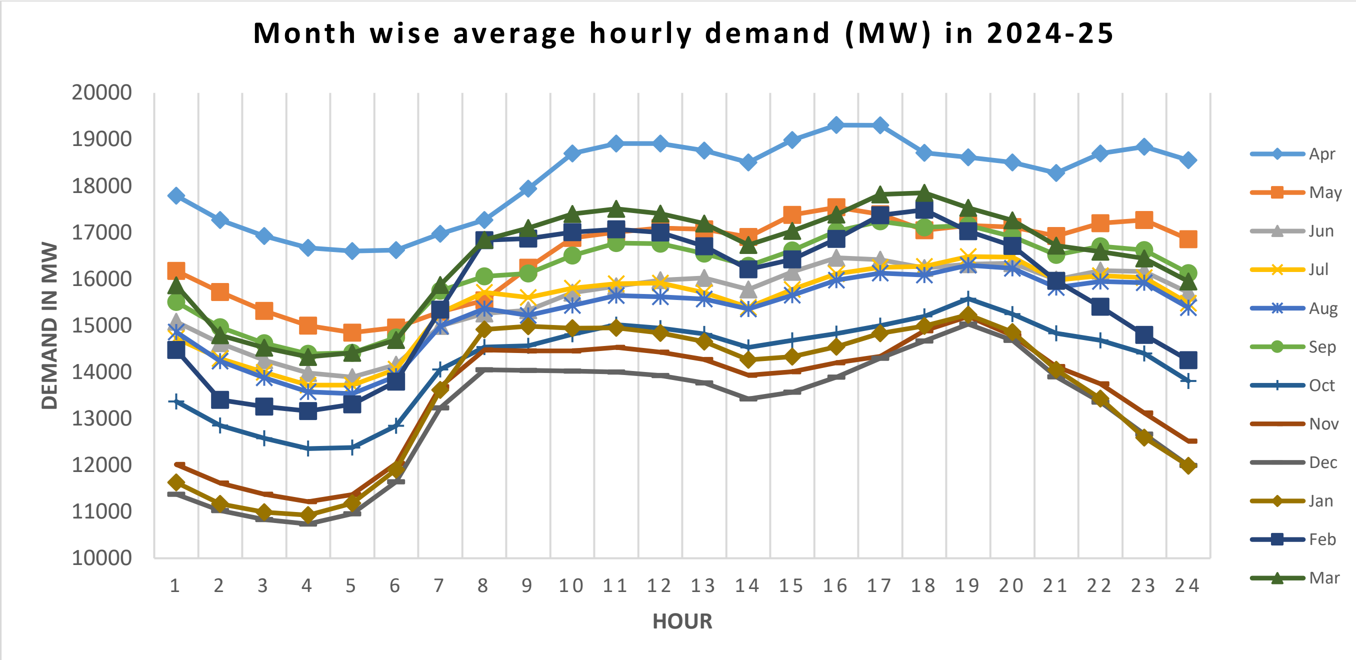


Figure 3: Average Hourly Demand Variation (Month-wise) for 2024-25

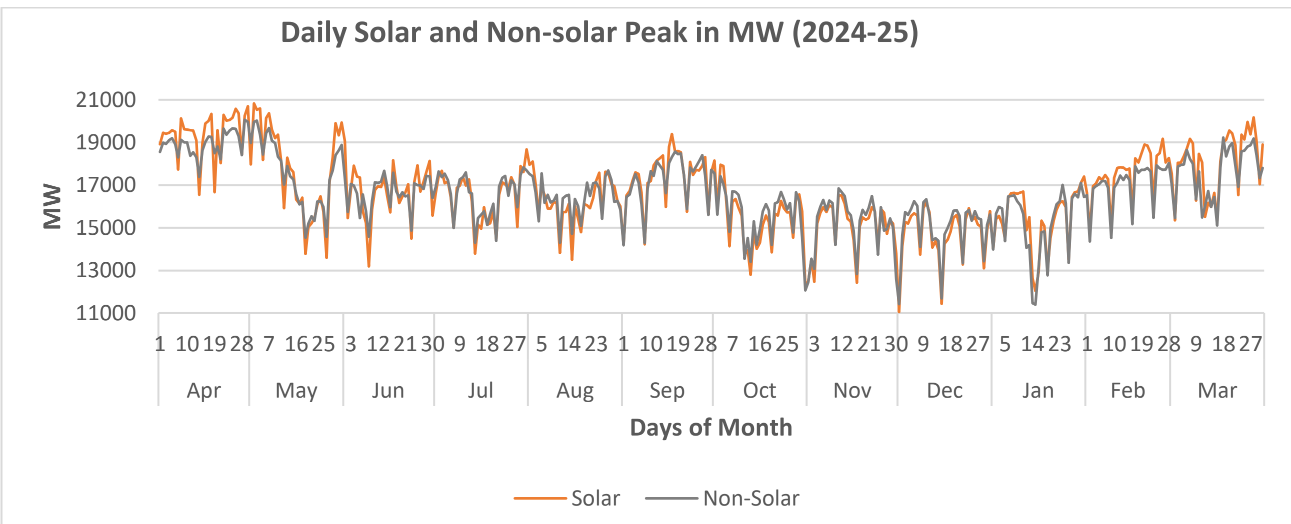


Figure 4: Solar and Non-Solar Peak Electricity Demand in MW for the year 2024-25

The hourly demand pattern of 2024-25 was analysed to find out the number of occurrences of the peak electricity demand and near-peak electricity demand. Such instances are critical for study purpose as it is necessary to ensure resource adequacy during such instances with an optimal mix of long-term, medium-term and short-term contracts.

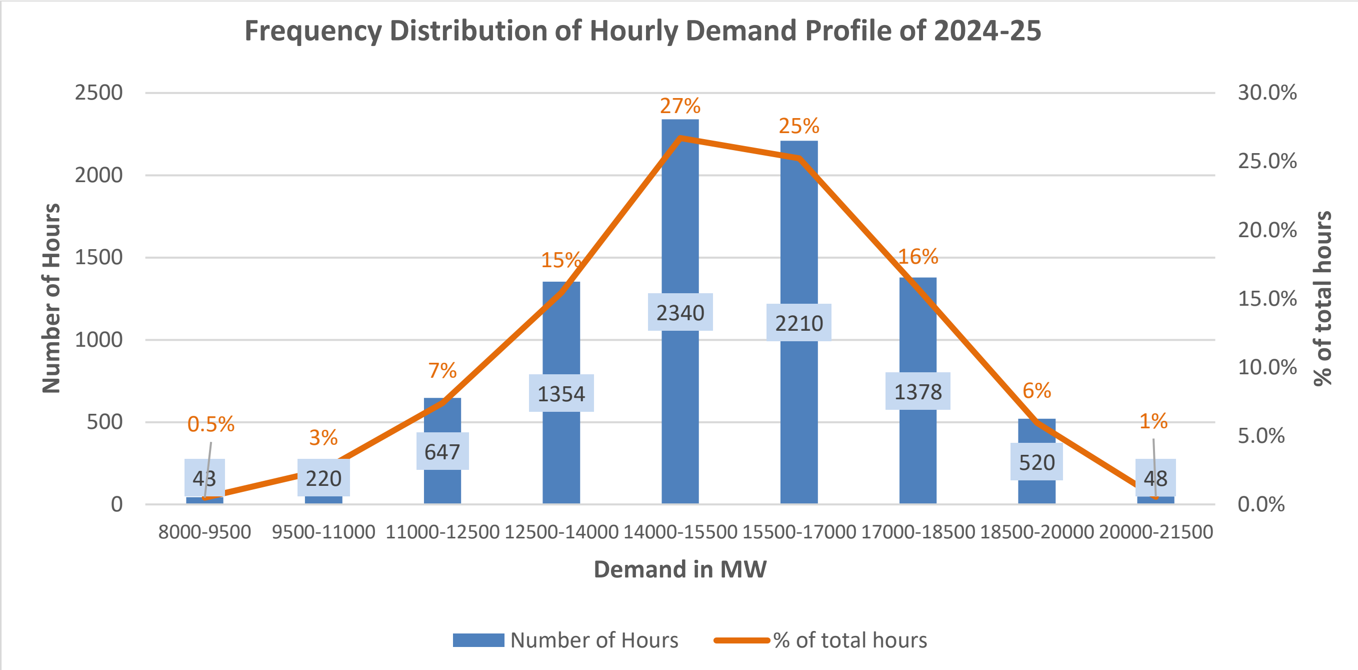


Figure 5: Frequency Distribution of Hourly Demand Profile of 2024-25

4.0 Principles of Generation Planning

The objective of Generation Planning is to determine the optimal cost-effective mix of generation capacities that can reliably meet electricity demand at all times while making the best use of available resources. Rather than attempting to predict the future, generation planning studies aim to collaboratively examine and understand the important factors and interrelationships that will influence the power sector moving forward. As such, planning is a continuous process focused on balancing demand and supply across all times and locations.

The major aspects considered in the planning process are:

i) To supply 24x7 reliable power to consumers
ii) To achieve the objectives of all policies of the Government of India, such as RCO trajectory, RE capacity addition, etc.
iii) To achieve sustainable development.
iv) To fulfil desired operational characteristics of the system, such as reliability and flexibility.
v) Most efficient use of resources.
vi) Fuel availability

4.1 Generation Expansion Planning Tool -ORDENA

The studies were carried out using a generation expansion planning model, namely ORDENA. ORDENA is a mixed integer linear optimisation program that minimises the Net Present Value (NPV) of investment and operation costs subject to several constraints. The major constraints include balancing electricity demand and supply, resource supply limits, planning and operating reserve limits, and policy targets. These constraints are met considering a broad portfolio of conventional generation, renewable generation, and storage.

ORDENA has a reliability module to determine the trustworthiness of the system using Monte Carlo simulations. The software is also capable of carrying out hourly/sub-hourly economic generation dispatch considering all the technical constraints associated with various generation technologies.

The schematic diagram of the software is given in the Exhibit.

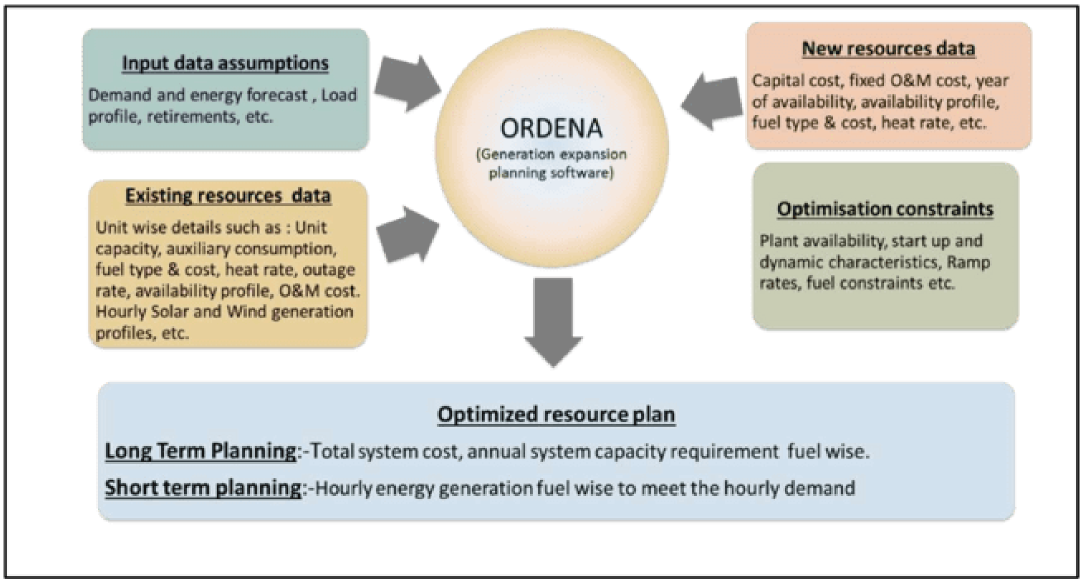


Exhibit: Schematic Diagram -Ordena

5.0 Reliability Analysis

One of the main criteria of resource adequacy studies is to determine the reliability of the system to meet the demand adequately at every instance in time. This reliability is measured via two indices (i.e.) LOLP (Loss of Load Probability) and EENS (Expected Energy Not Served). These indices have been defined in resource adequacy guidelines as below:

- **Loss of Load Probability (LOLP)**: Measure of the probability that a system's load may exceed the generation and firm power contracts available to meet that load in a year. E.g., 0.0274 % probability of load being lost.
- **Expected Energy Not Served (EENS):** Expected amount of energy (MWh) that may not be served for each year within the planning period under study. It is a summation of the expected number of megawatt hours of demand that may not be served for the year. This is an energy-centric metric that considers the magnitude and duration of energy not served,

calculated in megawatt-hours (MWh). The metric can be normalised (i.e., divided by the projected electrical energy requirement) to create a Normalised Energy Not Served (NENS) metric.

Monte Carlo /Stochastic simulation has been used to factor in the uncertainty associated with various generation resources and demand. It is an approach that is used to predict the probability of a variety of outcomes when the potential for random variables is present, as compared to deterministic modelling of the economic dispatch model. Monte Carlo simulation helps in analysing the randomness associated with RE energy resource, demand pattern changes and forced outages of the plant. A large number of random samples of these variables are simultaneously simulated to ascertain system reliability indices [i.e. Loss of Load Probability, LOLP and Energy Not Served, ENS] and the system robustness in case of the above variation of system parameters.

Planning Reserve Margin (PRM): To meet the prescribed standard of LOLP / NENS conditions, a sufficient reserve margin needs to be maintained in the system to adequately address the demand and supply variations. Planning Reserve Margin (PRM) is the predominant metric used to ensure adequacy of generation resources in the system. PRM in a power system is expressed as a certain percentage of the projected peak electricity demand of the system.

5.1 Variation in Electricity Demand

The hourly demand variation for consecutive years (i.e. 2023-24 and 2024-25, considering 2023-24 as the base) has been analysed. The demand pattern variation of 2023-24 and 2024-25 is shown below:

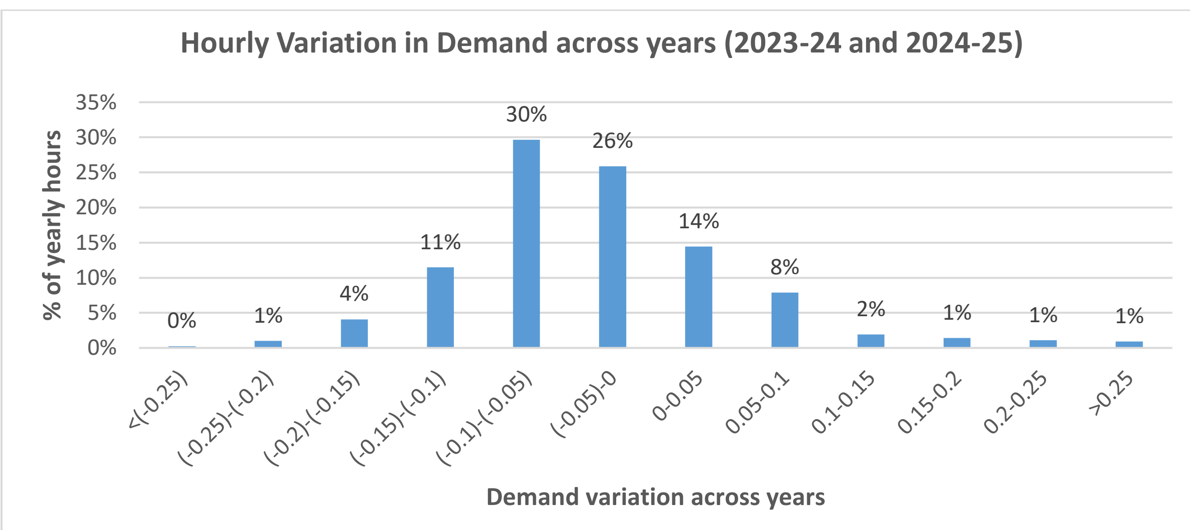


Figure 6: Hourly Variation in Demand across years

It can be observed that the hourly demand typically varies ±10% for around 78% of instances. This variation is primarily due to temperature, weather parameters or any random outages of transmission lines and generation units, etc. This variation has been captured in the reliability study by varying the projected hourly demand for the future years by ±10% by introducing a random variable (with normal distribution) for demand as per observed behaviour over the years.

5.2 Variation in RE

In the Long-term capacity expansion planning studies, a particular profile for Solar and wind is considered based on the observed solar generation data to determine the optimal capacity mix. However, due to the intermittent nature of these sources, the generation from these non-dispatchable sources may vary across years. As per the analyses carried out based on historical generation data, solar and hydro-based generation have been varied by ±10 % and wind generation has been varied by ±50% to incorporate the variation in these generation sources and plan for the requisite measures to mitigate such behavior.

5.3 Forced Outage of Thermal Generators

The average forced outage rate of thermal generators is typically 10% with ± 5% variation. The same has been incorporated in the model.

Based on these variations, reliability studies are carried out to ascertain the robustness of the system.

6.0 Inputs/Assumptions for the Study

i) The Electrical Energy Requirement (MU) and Electrical Peak Demand (MW) for the state of Tamil Nadu has been taken as per the projections by TNEB.

Table 7: Electricity Demand Projection as provided by TNEB

Electrical Energy Requirement (MU) and Peak Electricity Demand (MW) Projections				
	Energy Projections (MU)	Year-on-Year Growth	Peak Demand Projections (MW)	Year-on-Year Growth
2025-26	149063		21481	
2026-27	158408	5.9%	22750	6.3%
2027-28	168288	5.9%	24089	6.2%
2028-29	178720	5.9%	25502	6.2%
2029-30	190607	6.2%	27090	6.7%
2030-31	201053	5.4%	28542	5.5%
2031-32	212007	5.3%	30065	5.4%
2032-33	223884	5.5%	31707	5.6%
2033-34	236147	5.4%	33407	5.5%
2034-35	249652	5.6%	35263	5.7%
2035-36	262626	5.1%	37074	5.2%

ii) Future demand profile till the year 2035-36 has been projected using the demand profile for the year 2024-25 as the base profile.

iii) The actual solar and wind generation profiles and CUFs have been referred from the previous year's generation data provided by TNEB, while the hydro generation profiles and PLFs as per the data available in CEA.

iv) The capital costs of candidate plants for coal, wind, solar, and storage technologies is detailed in the Annexure and are aligned with current market trends and recent price discovery.

v) The planned capacity has been considered based on the tie-up information furnished by TNEB. Source-wise planned capacity is given below:

Table 8: Source-wise planned capacity addition

FY	Coal	Nuclear	Biomass	Hydro	Solar	Wind	Storage 2hr	Storage 4hr	Solar Rooftop DRE	PSP 6hr	PSP 8hr
2025-26	**0**	0	60	20	2100	300	0	0	500	0	0
2026-27	**3440**	500	15	14	3100	300	0	0	750	500	0
2027-28	**462**	652	0	0	1600	300	500	375	1250	0	0
2028-29	**819**	0	0	0	1600	300	0	700	1250	0	0
2029-30	**1500**	1000	0	0	1600	300	0	0	1750	0	0
2030-31	**0**	0	0	0	1600	300	0	0	1500	0	0
2031-32	**0**	0	0	0	1600	300	0	0	1100	1000	0
2032-33	**0**	0	0	0	1600	300	0	0	1100	0	1100
2033-34	**0**	0	0	0	1600	300	0	0	1100	1800	0
2034-35	**0**	0	0	0	1600	300	0	0	1100	1100	0
2035-36	**0**	0	0	0	1600	300	0	0	1100	0	0
Total	**6221**	**2152**	**75**	**34**	**19600**	**3300**	**500**	**1075**	**12500**	**4400**	**1100**

vi) **Renewable Consumption Obligation (RCO) trajectory**: Ministry of Power vide gazette notification dated 27th September 2025, had notified the source-wise minimum share of consumption of non-fossil sources (renewable energy) by designated consumers, till the year 2029-30.

Table 9: Renewable Consumption Obligation (RCO) trajectory as per the MoP order

Sl. No.	Year	Wind renewable energy	Hydro renewable energy	Other renewable energy	Distributed renewable energy	Total renewable energy
(1)	(2)	(3)	(4)	(5)	(6)	(7)
1.	2024-25	0.67%	0.38%	27.35%	1.5%	29.91%
2.	2025-26	1.45%	1.22%	28.24%	2.1%	33.01%
3.	2026-27	1.97%	1.34%	29.94%	2.7%	35.95%
4.	2027-28	2.45%	1.42%	31.64%	3.3%	38.81%
5.	2028-29	2.95%	1.42%	33.1%	3.9%	41.36%
6.	2029-30	3.48%	1.33%	34.02%	4.5%	43.33%

Further, in view of the country's energy transition goals as well as the long-term net zero target of 2070, it is estimated that the share of RE generation in the generation mix will continue to proportionally increase beyond 2029-30. Therefore, the RCO trajectory has been assumed to rise steadily beyond 2029-30. Hence, the suggested trajectory of Renewable Consumption Requirement up to FY 2035-36 is given in Table 10.

Table 10: Renewable Consumption Obligation (RCO) trajectory considered for the study

<table>
<tr><th>Sl. No.</th><th>Year</th><th>Wind renewable energy</th><th>Hydro renewable energy</th><th>Other renewable energy</th><th>Distributed renewable energy</th><th>Total renewable energy</th></tr>
<tr><td>1.</td><td>2024-25</td><td>0.67%</td><td>0.38%</td><td>27.35%</td><td>1.5%</td><td>29.91%</td></tr>
<tr><td>2.</td><td>2025-26</td><td>1.45%</td><td>1.22%</td><td>28.24%</td><td>2.1%</td><td>33.01%</td></tr>
<tr><td>3.</td><td>2026-27</td><td>1.97%</td><td>1.34%</td><td>29.94%</td><td>2.7%</td><td>35.95%</td></tr>
<tr><td>4.</td><td>2027-28</td><td>2.45%</td><td>1.42%</td><td>31.64%</td><td>3.3%</td><td>38.81%</td></tr>
<tr><td>5.</td><td>2028-29</td><td>2.95%</td><td>1.42%</td><td>33.1%</td><td>3.9%</td><td>41.36%</td></tr>
<tr><td>6.</td><td>2029-30</td><td>3.48%</td><td>1.33%</td><td>34.02%</td><td>4.5%</td><td>43.33%</td></tr>
<tr><td>7.</td><td>2030-31</td><td colspan="3">40.50%</td><td>5.0%</td><td>45.5%</td></tr>
<tr><td>8.</td><td>2031-32</td><td colspan="3">41.50%</td><td>5.5%</td><td>47.0%</td></tr>
<tr><td>9.</td><td>2032-33</td><td colspan="3">42.30%</td><td>6.0%</td><td>48.3%</td></tr>
<tr><td>10.</td><td>2033-34</td><td colspan="3">43.00%</td><td>6.5%</td><td>49.5%</td></tr>
<tr><td>11.</td><td>2034-35</td><td colspan="3">43.50%</td><td>7.0%</td><td>50.5%</td></tr>
<tr><td>12.</td><td>2035-36</td><td colspan="3">44.00%</td><td>7.5%</td><td>51.5%</td></tr>
</table>

As per Ministry of Power vide notification dated 27th September 2025, had notified the RCO trajectory for Designated Consumers/Discoms, For all the designated consumers, the Renewable Consumption Obligation shall exclude electricity consumed from Nuclear Power Sources. The Electrical Energy Requirement (MU) after excluding Nuclear Power Sources is tabulated below:

Table 11: Total Energy required to meet RCO (MU)

	Electrical Energy Requirement (MU)	Nuclear Energy (MU)	Adjusted Energy Requirement (MU)
2025-26	149063	7763	141300
2026-27	158408	10443	147965
2027-28	168288	13938	154350
2028-29	178720	8979	169741
2029-30	190607	14340	176267
2030-31	201053	14340	186713
2031-32	212007	14340	197667
2032-33	223884	14340	209544
2033-34	236147	14340	221807
2034-35	249652	14340	235312
2035-36	262626	14340	248286

Based on the trajectory specified, RCO quantum in million units (MUs) from hydro, wind, other (Solar, biomass, etc.) and Distributed Renewable Energy (DRE) has been calculated and is tabulated below.

Table 12: Total Energy required to meet RCO (MU)

Sl. No.	Year	Wind renewable energy (MU)	Hydro renewable energy	Other renewable energy	Distributed renewable energy	Total renewable energy
1.	2025-26	2049	1724	39903	2967	46643
2.	2026-27	2915	1983	44301	3995	53193
3.	2027-28	3782	2192	48836	5094	59903
4.	2028-29	5007	2410	56184	6620	70205
5.	2029-30	6134	2344	59966	7932	76376
6.	2030-31	75619			9336	84955
7.	2031-32	82032			10872	92903
8.	2032-33	88637			12573	101210
9.	2033-34	95377			14417	109795
10.	2034-35	102361			16472	118833
11.	2035-36	109246			18621	127867

The year-wise total renewable energy, excluding DRE, met from the existing and planned capacity along with the deficit is shown in Table 13.

Table 13 Renewable Energy Deficit/Surplus

FY	RE Generation required to meet RCO (excluding DRE)		RE Generation (excluding DRE) available or to be met from existing and planned contracts		RE Generation Surplus (+)/ Deficit (-) (excluding DRE)
	(MU)	**(%)**	**(MU)**	**(%)**	**(%)**
2025-26	43676	30.91%	38758	27.43%	-3.48%
2026-27	49198	33.25%	45514	30.76%	-2.49%
2027-28	54810	35.51%	49350	31.97%	-3.54%
2028-29	63585	37.46%	53186	31.33%	-6.13%
2029-30	68444	38.83%	57022	32.35%	-6.48%
2030-31	75619	40.50%	60858	32.59%	-7.91%
2031-32	82032	41.50%	64694	32.73%	-8.77%
2032-33	88637	42.30%	68530	32.70%	-9.60%
2033-34	95377	43.00%	72366	32.63%	-10.37%
2034-35	102361	43.50%	76202	32.38%	-11.12%
2035-36	109246	44.00%	80038	32.24%	-11.76%

Similarly, the year-wise total renewable energy for Distributed Renewable Energy (DRE) met from the existing and planned capacity along with the deficit is shown in Table below:

Table 14 Distributed Renewable Energy (DRE) Deficit/Surplus

FY	DRE Generation required to meet RCO		DRE generation available or met from existing and planned contracts		DRE Generation Surplus (+)/ Deficit (-)
	(MU)	(%)	(MU)	(%)	(%)
2025-26	2967	2.10%	1991	1.41%	-0.69%
2026-27	3995	2.70%	2941	1.99%	-0.71%
2027-28	5094	3.30%	4525	2.93%	-0.37%
2028-29	6620	3.90%	6109	3.60%	-0.30%
2029-30	7932	4.50%	8326	4.72%	0.22%
2030-31	9336	5.00%	10226	5.48%	0.48%
2031-32	10872	5.50%	11620	5.88%	0.38%
2032-33	12573	6.00%	13014	6.21%	0.21%
2033-34	14417	6.50%	14407	6.50%	0.00%
2034-35	16472	7.00%	15801	6.71%	-0.29%
2035-36	18621	7.50%	17195	6.93%	-0.57%

As indicated in the tables above, the state will not be able to meet its RCO requirements based on the existing and planned contracts from 2025-26 to 2035-36.

7.0 Outcome of the model

7.1 Unserved Electrical Energy Projections

Initially, the study was carried out considering only the existing and planned capacity contracts by Tamil Nadu without considering any banking arrangements or short-term contracts. The total unserved electrical energy for the year 2035-36 would be 56,294 MU in this case. Year-wise likely unserved electrical energy with the existing and planned capacities is given in Figure 7.

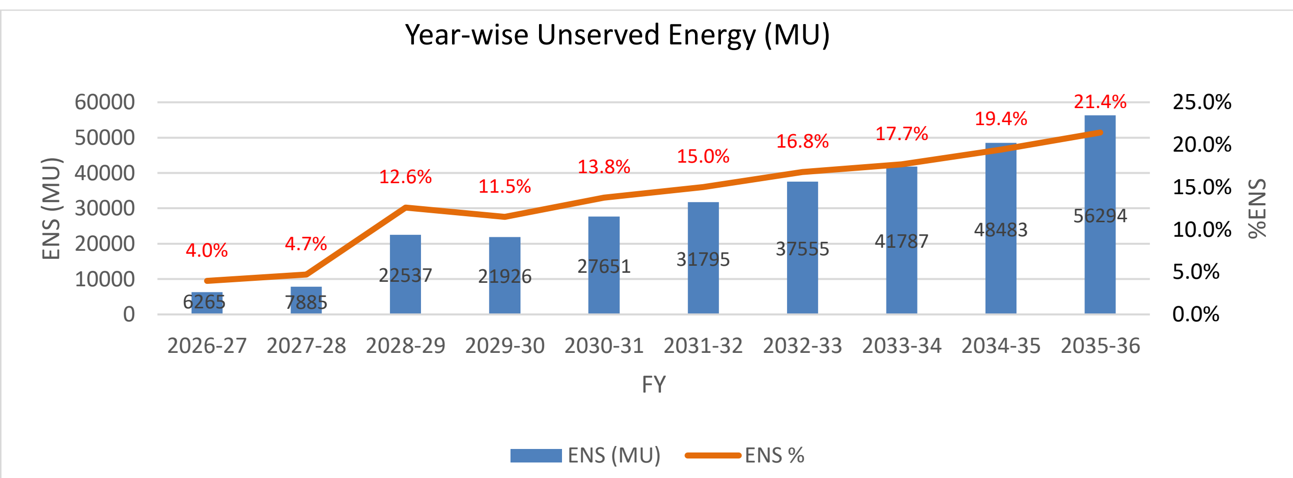


Figure 7: Yearly likely unserved electrical energy (in MU) with the existing and planned capacities

The study has also analyzed the daily and monthly pattern of unserved energy in the year 2035-36 and it can be seen that contracted capacity (existing and planned) is unable to meet the projected electricity demand.

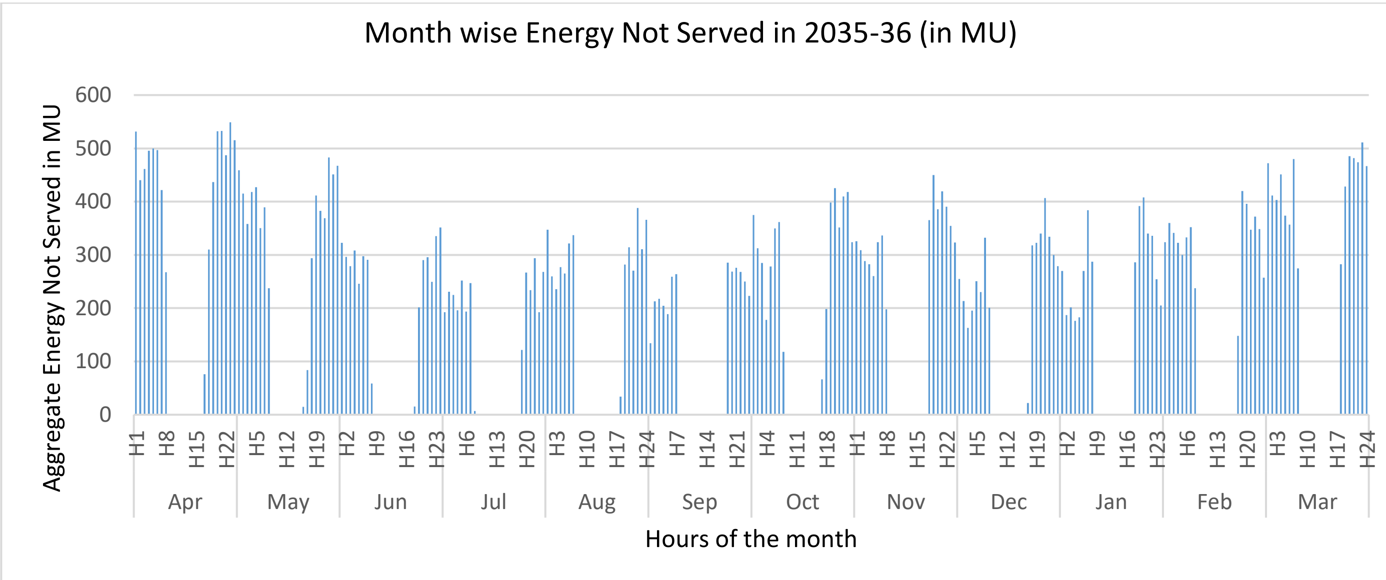


Figure 8: Block-wise Unserved Electrical Energy Pattern MU (2035-36)

The unmet demand occurs mainly during non-solar hours throughout the year. To meet unserved energy, the utility needs to tie up additional capacity.

7.2 Options to meet the unserved electrical energy

To meet this unserved energy, options (i.e., candidate capacities) have been given to the model to find the least cost optimal capacity mix required to meet the projected electricity demand with RCO obligations. The capacity projections for Tamil Nadu are given below in Table 15 and Figure 9:

Table 15: Likely requirement of contracted capacity (existing + planned + additional) (in MW)

Year	Coal	Gas	Nuclear	Biomass	Hydro	Wind	Solar	PSP	Storage	Solar DRE	Hydro DRE	STOA
2025-26	12808	408	1448	951	1886	9631	11106	400	0	2353	58	6727
2026-27	16248	408	1948	966	1900	11181	15456	900	0	3103	58	4470
2027-28	16265	408	2600	966	1900	12181	17756	900	2875	4353	58	4360
2028-29	15962	408	1675	966	1900	13181	20056	900	5575	5603	58	5573
2029-30	17754	408	2675	966	1900	14181	22356	1144	7575	7353	58	4147
2030-31	18754	408	2675	966	1900	15181	24656	1144	9575	8853	58	3959
2031-32	19754	408	2675	966	1900	16181	26956	2144	11233	9953	58	3529
2032-33	20504	408	2675	966	1900	17181	29256	3244	11363	11053	58	3642
2033-34	21504	408	2675	966	1900	18181	31556	5044	11363	12153	58	3208
2034-35	22504	408	2675	966	1900	19181	33856	6144	11575	13303	58	3375
2035-36	23504	408	2675	966	1900	20181	36156	6264	11575	14703	58	4107

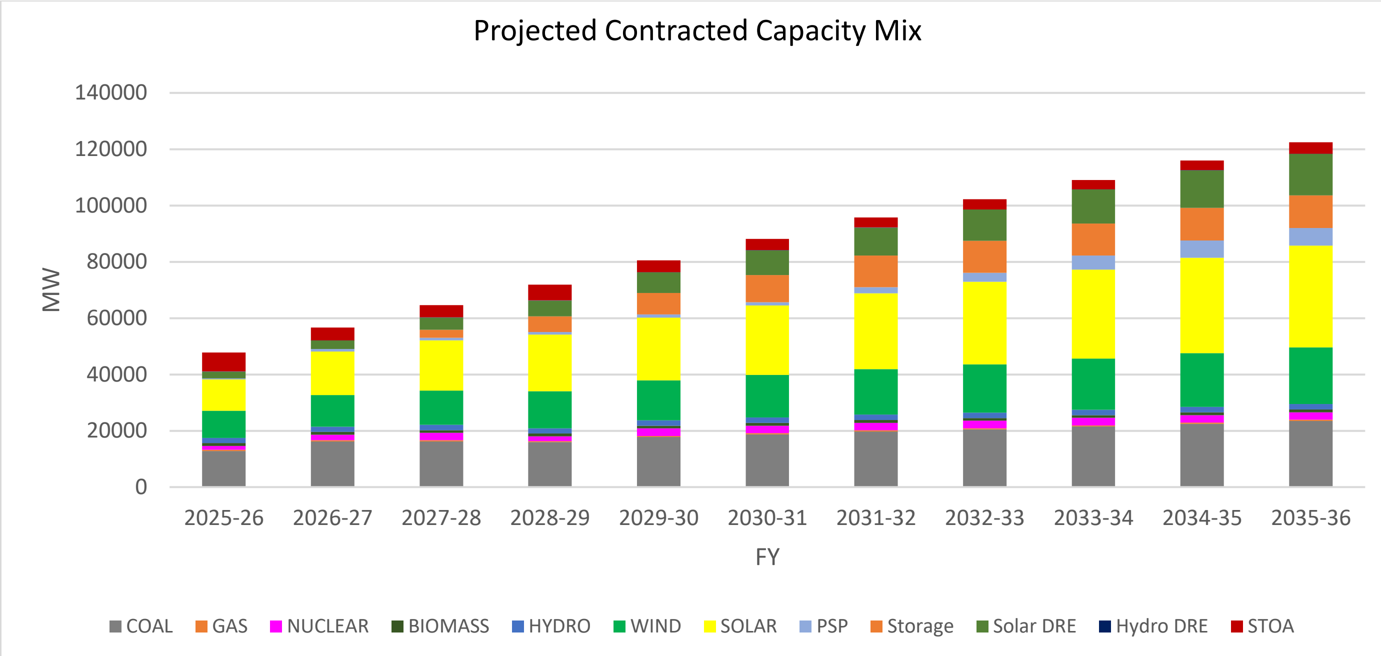


Figure 9: Projected Contracted Capacity Mix Year-wise (MW)

As per the study, the utility needs to contract the following capacities (planned and additional) per year till 2035-36 to reliably meet its electricity demand.

Table 16: Likely planned and additional capacity to be contracted by the utility (in MW)

	Coal		Hydro	Nuclear	Biomass	Solar		Wind	
	Planned	**Additional**	**Planned**	**Planned**	**Planned**	**Planned**	**Additional**	**Planned**	**Additional**
2025-26	0	0	20	0	60	2100	0	300	0
2026-27	3440	0	14	500	15	3100	1250	300	1250
2027-28	462	0	0	652	0	1600	700	300	700
2028-29	819	1000	0	0	0	1600	700	300	700
2029-30	1500	1000	0	1000	0	1600	700	300	700
2030-31	0	1000	0	0	0	1600	700	300	700
2031-32	0	1000	0	0	0	1600	700	300	700
2032-33	0	1000	0	0	0	1600	700	300	700
2033-34	0	1000	0	0	0	1600	700	300	700
2034-35	0	1000	0	0	0	1600	700	300	700
2035-36	0	1000	0	0	0	1600	700	300	700
Total	**6221**	**8000**	**34**	**2152**	**75**	**19600**	**7550**	**3300**	**7550**

Year	Storage			PSP			DRE	
	Planned Storage 2h	**Planned Storage 4h**	**Additional Storage 4h**	**Planned PSP 6h**	**Planned PSP 8 h**	**Additional PSP 6h**	**Planned**	**Addition al**
2025-26	0	0	0	0	0	0	500	850
2026-27	0	0	0	500	0	0	750	0
2027-28	500	375	2000	0	0	0	1250	0
2028-29	0	700	2000	0	0	0	1250	0
2029-30	0	0	2000	0	0	244	1750	0
2030-31	0	0	2000	0	0	0	1500	0
2031-32	0	0	1658	1000	0	0	1100	0
2032-33	0	0	130	0	1100	0	1100	0
2033-34	0	0	0	1800	0	0	1100	0
2034-35	0	0	212	1100	0	0	1100	50
2035-36	0	0	0	0	0	120	1100	300
Total	**500**	**1075**	**10000**	**4400**	**1100**	**364**	**12500**	**1200**

The STOA/MTOA requirement can be fulfilled through power procurement from markets or bilateral agreements. The STOA/MTOA value reflects the peak value requirement/seasonal requirements in terms of MW.

Reliability studies have been conducted based on the projected capacity for the year 2035-36. The analysis confirms that the projected capacity meets the reliability criteria specified in the National Electricity Plan (NEP), with the Loss of Load Probability (LoLP) and Not-Served Energy (NENS) remaining within the permissible limits of 0.2% and 0.05%, respectively as specified in the National Electricity Plan (Generation).

As per the Resource Adequacy studies, the likely total projected contracted capacity for the year 2035-36 is around 1,22,498 MW which consists of 23,504 MW from Coal; 408 MW from Gas; 2675 MW from Nuclear; 1900 MW from Hydro; 20,181 MW from Wind; 36,156 MW from Solar; 14,761 MW from Distributed Renewable Energy (DRE) sources; 966 MW from Biomass; 11,575 MW/45300 MWh of Storage; 6264 MW/39782 MWh of PSP and 4107 MW of MTOA/STOA. This capacity shall be able to meet the projected electricity demand with prescribed reliability criteria.

It is important to note that any deviations in the commissioning schedule of the planned capacity could result in a situation where the utility is unable to meet the projected peak electricity demand and electrical energy requirements identified in this study with the available resources. Such changes may also lead to an increase in the cost of reliably meeting the utility’s power demand.

The Planning Reserve Margin (PRM) for Tamil Nadu has been assessed as 4%. Furthermore, the projected/contracted capacity is in compliance with the stipulated Renewable Consumption Obligation (RCO) targets.

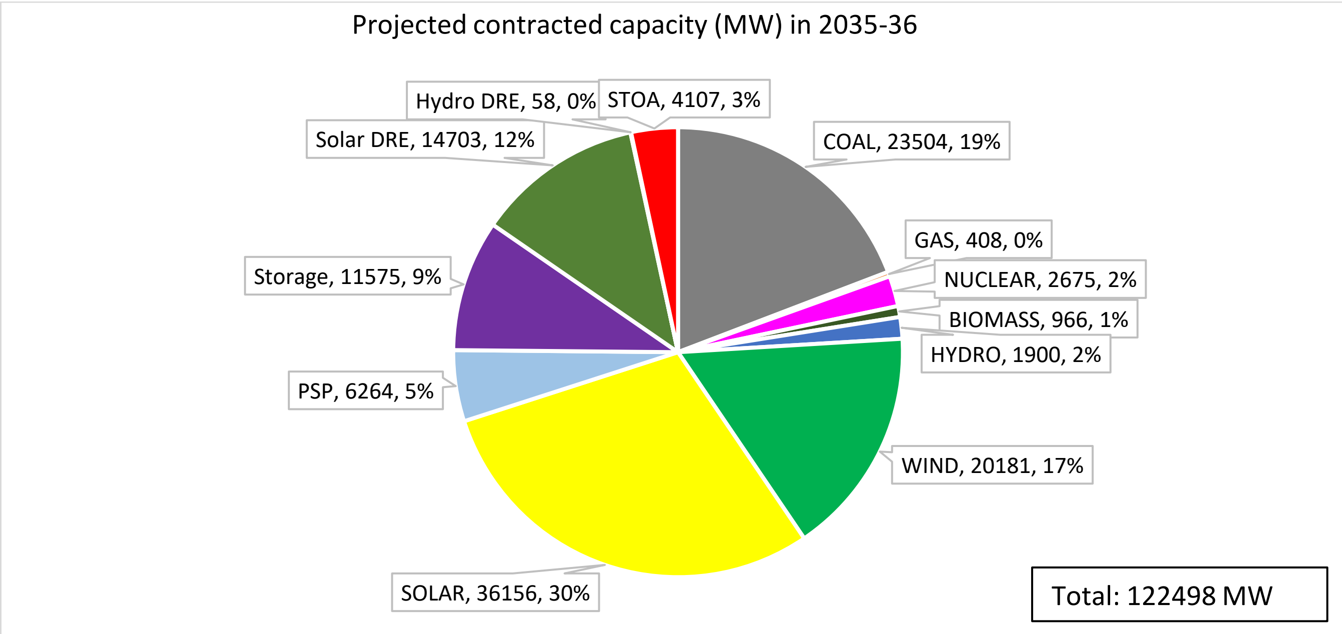


Figure 11: Contracted Capacity Mix in 2035-36

The share of non-fossil fuel-based capacity is projected to increase to around 78% by 2035-36 from 64% in 2024-25. Further, the share of non-fossil fuel-based generation will be around 59% in the year 2035-36. The projected generation mix for the utility is shown in Figure 12.

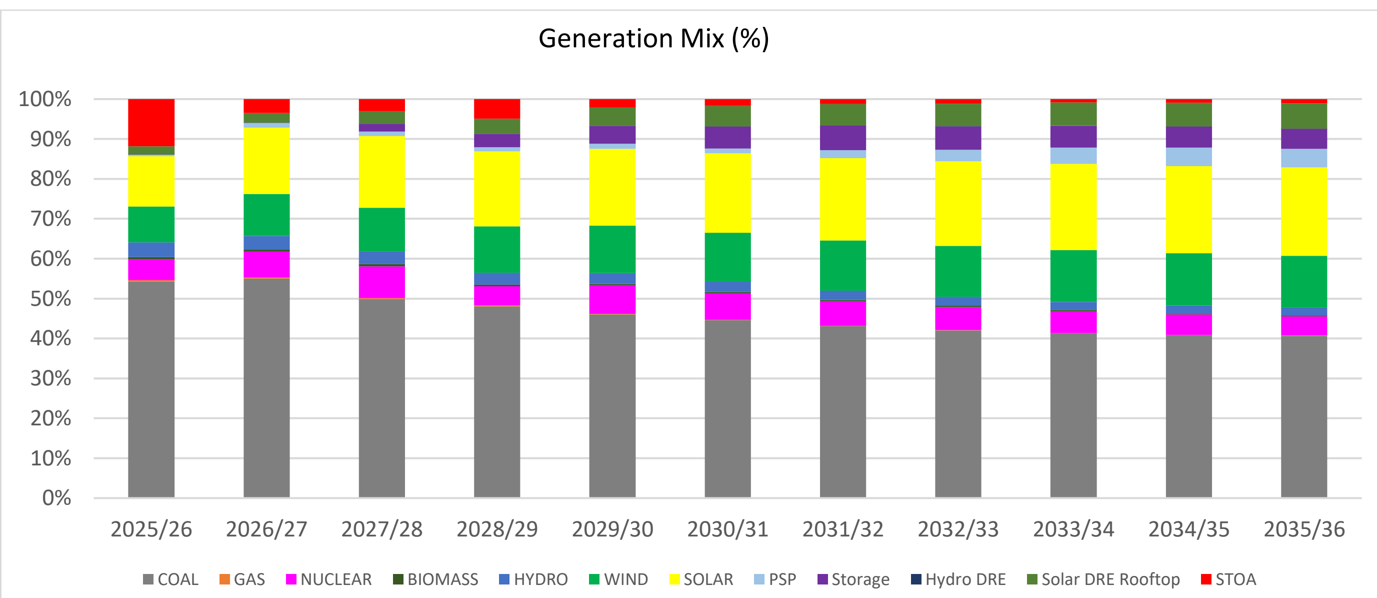
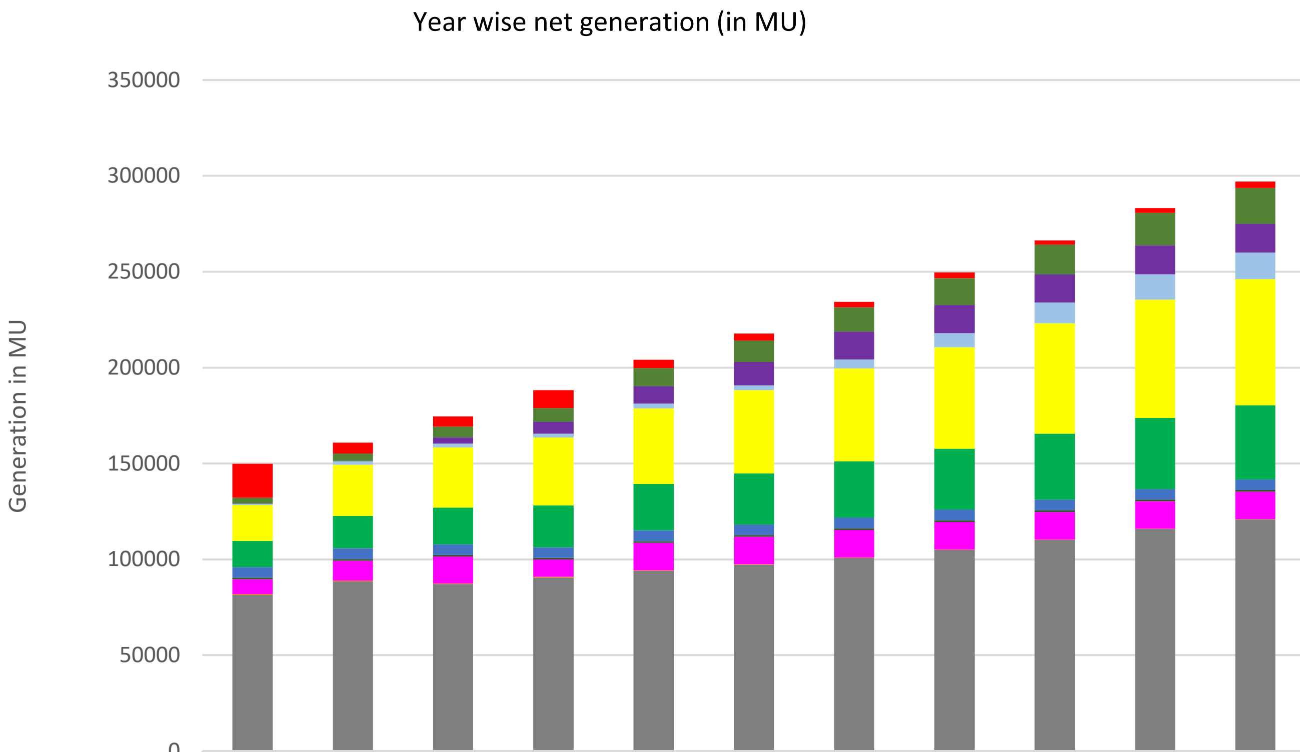


Figure 12: Year-wise projected generation mix (in %)

The year-wise projected generation in MU is depicted below:



	2025/26	2026/27	2027/28	2028/29	2029/30	2030/31	2031/32	2032/33	2033/34	2034/35	2035/36
STOA	17752	5597	5305	9292	4388	3706	2801	2947	2299	2503	3279
Solar DRE Rooftop	2982	3932	5516	7100	9317	11217	12611	14005	15399	16856	18629
Hydro DRE	86	86	86	86	86	86	86	86	86	86	86
Storage	0	0	3285	6205	9125	12045	14466	14655	14655	14965	14965
PSP	618	1963	1968	1964	2505	2498	4633	7308	10899	13200	13924
SOLAR	18844	26677	31303	35404	39427	43469	48426	52954	57465	61845	65725
WIND	13544	16894	19233	21899	24196	26726	29470	31801	34535	37042	38775
HYDRO	5528	5549	5549	5549	5549	5549	5549	5549	5549	5549	5549
BIOMASS	842	857	857	857	857	857	857	857	857	857	857
NUCLEAR	7763	10443	13938	8979	14340	14340	14340	14340	14340	14340	14340
GAS	521	396	368	413	342	270	181	184	149	137	183
COAL	81356	88468	87073	90479	93976	97100	100816	104987	110192	115777	120725

COAL GAS NUCLEAR BIOMASS HYDRO WIND SOLAR PSP Storage Hydro DRE Solar DRE Rooftop STOA

Figure 13: Year-wise projected net generation (in GWh)

7.3 Day-wise Surplus Capacity (in MW)

The pattern of surplus capacities has been observed as given in Figure 14. Available Surplus capacity can be due to seasonal variation in demand. This capacity can be used for banking arrangements or shared with other states through bilateral agreements / Pushp portal, thereby reducing the fixed cost burden on the utility, resulting in a reduction in the cost for the consumer. Tamil Nadu is likely to have surplus capacity during the off-peak electricity demand months (July to September), which can be shared with other states. Specifically, the utility is likely to have a minimum surplus of around 2000 MW in the month of July 2028-29, which can be shared with other states/Discoms.

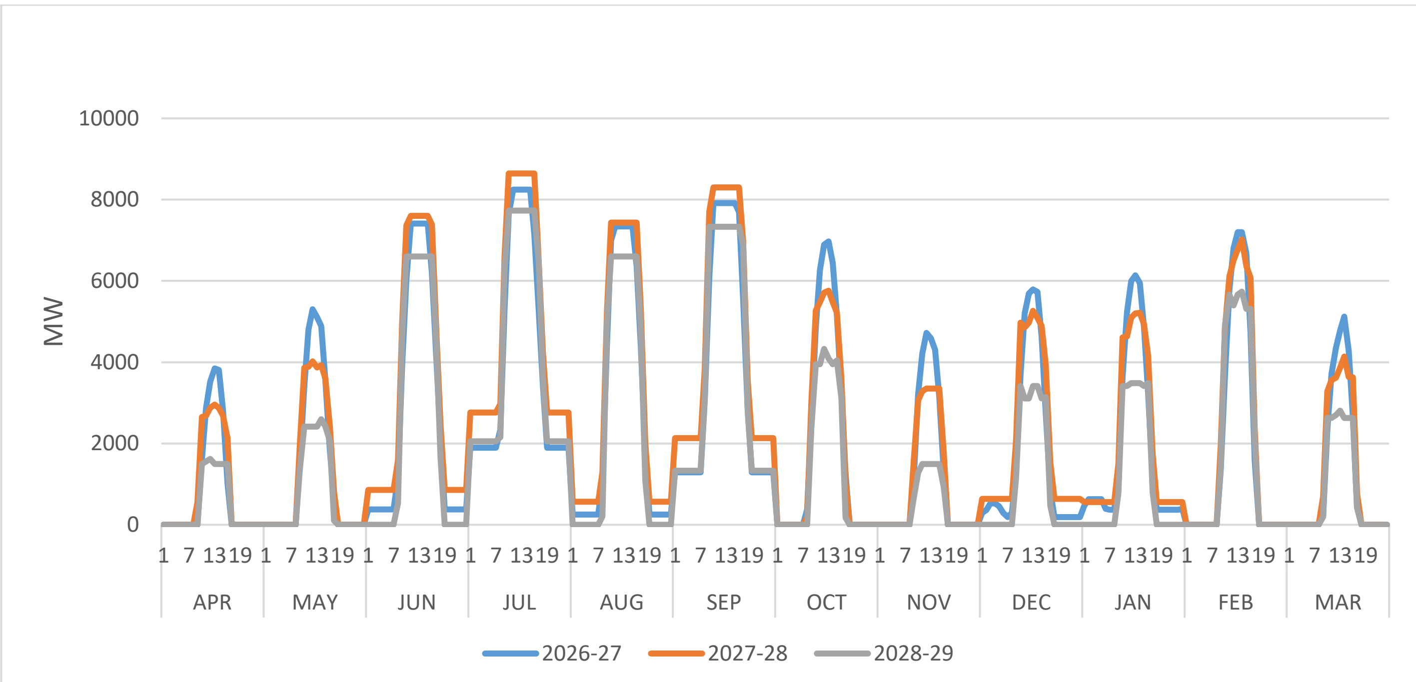


Figure 14: Surplus Coal Capacity Year-wise (MW)

8.0 Conclusions

Based on the Resource Adequacy studies of Tamil Nadu up to the year 2035-36, the following conclusions may be drawn:

1. Based on the demand pattern of the last few years, it is seen that the high electricity demand is typically in the months of March and April, with peak electricity demand occurring during solar hour. The demand during the months from July to September remains significantly low compared to other months, reflecting the seasonality in demand. Optimal utilisation of resources through short-term contracts like banking or MTOA/STOA for managing the seasonal variation in demand is one of the effective ways for ensuring resource adequacy in such periods.

2. The study has considered electrical energy requirement projections and peak electricity demand projections as prepared by TNEB which envisages that the annual electrical energy requirement of the utility is likely to grow at a CAGR of 5.83% while the peak electricity demand for the utility is likely to grow at a CAGR of 5.61% for the period of 2025-26 to 2035-36.

3. Tamil Nadu is likely to witness energy deficit ranging from 6265 MUs to 56294 MUs in different years from 2025-26 to 2035-36 with the existing and planned capacity addition only. The deficit may reduce depending upon the banking or MTOA/STOA arrangement, etc.

4. As per the Resource Adequacy studies, the likely total projected contracted capacity for the year 2035-36 is around 1,22,498 MW which consists of 23,504 MW from Coal; 408 MW from Gas; 2675 MW from Nuclear; 1900 MW from Hydro; 20,181 MW from Wind; 36,156 MW from Solar; 14,761 MW from Distributed Renewable Energy (DRE) sources; 966 MW from Biomass; 11,575 MW/45300 MWh of Storage; 6264 MW/39782 MWh of PSP and 4107 MW of MTOA/STOA. This capacity shall be able to meet the projected electricity demand with prescribed reliability criteria. The Storage requirement can be met through PSP or a combination of PSP and BESS.

5. To ensure reliability of supply and adherence to the Renewable Consumption Obligation (RCO) trajectory up to 2035–36, additional capacity, over and above the planned capacity, needs to be tied up by the State. An additional capacity of approximately 8000 MW of coal-based power, 7550 MW of solar and 7550 MW of wind is projected to be required. In addition to this, the utility will likely require MTOA/ STOA arrangements ranging from around 3000 MW to 7000 MW in different years, as outlined in the report. The STOA/MTOA requirement can be fulfilled through power procurement from the market or bilateral/banking agreements.

6. The current capacity mix of the utility has 36% of contracted capacity from fossil fuel sources (Coal + Gas). The share of fossil fuel-based capacity is likely to reduce to 22% in 2035-36. In

2035-36, the RE-based contract capacity (that includes hydro, solar, wind) is likely to constitute around 64% of the total contract capacity.

7. It is likely that Tamil Nadu may have surplus capacity available during the months from July to September, which can be shared with other utilities/states through banking/bilateral arrangements, including the Pushp portal.
8. The Planning Reserve Margin (PRM) for Tamil Nadu has been assessed as 4%. Further, the study indicates year-wise short-term/medium-term/bilateral requirements to meet the demand optimally.
9. Accurate demand assessment is fundamental to capacity expansion modelling and directly impacts the Resource Adequacy (RA) study outcomes. To achieve realistic and data-driven results, utilities are strongly encouraged to adopt scientific tools and robust methodologies for demand forecasting.
10. Timely commissioning of planned capacities is critical to ensure that the utility can meet the projected peak electricity demand and electrical energy requirements identified in this study. Any deviation from the planned schedule may lead to resource shortfalls and could significantly increase the cost of reliably meeting the utility's power demand.

Annexure

Future Contracted/Approved Capacity Expansion and Termination (MW)

Sl. No.	Name of Power Plant	Type of Generation	Expected CoD	Capacity tie up/share (MW)
1	Ennore SCTPP / TANGEDCO U-1	Coal	2026-27	660
2	Ennore SCTPP / TANGEDCO U-2	Coal	2026-27	660
3	Udangudi STPP Unit 1	Coal	2026-27	660
4	Udangudi STPP Unit 2	Coal	2026-27	660
5	North Chennai TPP St-III	Coal	2026-27	800
6	TALCHER TPS-III	Coal	2027-28	462
7	NLC TPS II 2nd Expansion	Coal	2028-29	819
8	NLC Talabira STPS-Odisha U1-2	Coal	2029-30	1000
9	NLC Talabira STPS-Odisha U3	Coal	2029-30	500
10	Kudankulam Unit 3	Nuclear	2026-27	500
11	Kudankulam Unit 4	Nuclear	2027-28	500
12	PFBR NEW	Nuclear	2027-28	152
13	Kudankulam Unit 5	Nuclear	2029-30	500
14	Kudankulam Unit 6	Nuclear	2029-30	500
15	Kolli Small Hydro	Hydro	2025-26	20
16	RMU of Kodayar and Moyar	Hydro	2026-27	14
17	Kundah PSP	PSP	2026-27	500
18	Upper Bhavani PSP	PSP	2031-32	1000
19	Vellimalai PSP	PSP	2032-33	1100
20	Aliyar PSP	PSP	2033-34	1800
21	Palar-Poranthalar PSP	PSP	2034-35	1100

Sl. No.	Name of Power Plant	Type of Generation	Expected Year of PPA Termination/Retirement	Capacity tied up (MW)
1	GMRKEL	Coal	2027-28	102
2	NEYVELI (EXT) TPS Unit 1	Coal	2027-28	193
3	NEYVELI TPS- II	Coal	2027-28	149
4	ADHUNIK	Coal	2028-29	100
5	BALCO-I (LTOA)	Coal	2028-29	100
6	BALCO-II (LTOA)	Coal	2028-29	100
7	COASTAL	Coal	2028-29	558
8	DHARIWAL TPP	Coal	2028-29	100
9	GMR	Coal	2028-29	150
10	IL&FS	Coal	2028-29	540
11	Jindal (LTOA)	Coal	2028-29	400
12	OPG (LTOA)	Coal	2028-29	74
13	DB Power (LTOA)	Coal	2029-30	208
14	KSK	Coal	2029-30	500
15	TAQA Neyveli Power Company Private Limited	Coal	2032-33	250
16	Kudankulam Unit 1&2	Nuclear	2028-29	925

Assumption for Resource Adequacy Studies

1. Electrical energy requirement & peak electricity demand requirement: As per projections of Tamil Nadu.
2. Demand Profile: Based on the hourly demand profile of 2024-25.
3. Existing & Planned Capacity: As per the information shared by the utility.
4. Cost parameters: based on information in the National Electricity Plan, recent market trends, and existing tariff information received from TNEB.

RE CUF considered

Hydro	Solar CUF Existing & Planned	Wind CUF Existing & Planned	DRE
32 %	18.8% & 22%	15.6% & 29%	14.5%

Technical Parameters

Technology	Type	Availability (%)	Ramping (%/min)	Min. Technical (%)
Coal/ Lignite	Existing/Planned	85	1	55
	Candidate	88	1	55
Gas	Existing	90	5	40
Nuclear	Existing/Planned	68	Const. Load	-
Biomass	Existing/Planned	60	2	50
Hydro	Existing/Planned/ Candidate	As per available hourly generation profile	100	-
Solar	Existing/Planned		-	-
	Candidate		-	-
Wind	Existing/Planned		-	-
	Candidate		-	-
Pumped storage	Existing/Planned	95	50	-
	Candidate		50	-
Battery Energy Storage	Candidate	98	NA	-

Technology	Type	Heat Rate(MCal/MWh)	Aux. Consum. (%)
Coal	Existing/ Planned	2104 to 2569	7.0
	Candidate (SC & USC)	2200	6.5
Gas	Existing	2000 to 2900	2.5
Nuclear	Existing/ Planned	2777	10
	Candidate	2777	10
Biomass	Existing/ Planned	4200	8
Hydro	Existing/ Planned	-	0.7
	Candidate	-	0.7
Pumped Storage Project	Existing/ Planned	-	Round trip efficiency efficiency 80%
	Candidate	-	
Battery Energy Storage System	Candidate	-	Round trip efficiency 88%

Transmission Parameters

A single node has been considered for the purpose of study with all generating units and demand connected to the node. No transmission bottleneck has been considered for the study. Interstate ATC limit has not been considered in the study.

Financial Parameters

The following cost parameters have been assumed:

Resource	Capex (in ₹/MW)	O&M Fixed Cost (in ₹/MW)	Construction Time (in years)	Amortization /Lifetime (in years)
Coal	12 Cr	19.54 Lakh	4	25
Solar	4.4 Cr to 4.1 Cr	1 % of Capex	1	25
Wind	7.5 Cr	1 % of Capex	1.5	25
Battery Energy Storage (2-Hour)	3.21 Cr to 2.63 Cr	1 % of Capex	1	14
Battery Energy Storage (4-Hour)	4.98 Cr to 3 Cr	1 % of Capex	1	14
Pumped Storage	6 Cr	5 % of Capex	5	40